Exercises 4

Exercise 1 (The quantum SWAP gate)**.** *The two-qubit* SWAP *gate is defined on the standard basis states as follows:*

$$\text{SWAP}\,|a,b\rangle = |b,a\rangle, \tag{1}$$

for all $a, b \in \{0,1\}$*, where* $|a,b\rangle$ *is a shorthand for* $|a\rangle \otimes |b\rangle$*. This definition is then extended by linearity to all two-qubit states.*

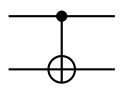
1. *Write down the matrix representation of* SWAP*.*
2. *Let* $|\alpha\rangle = \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ *and* $|\beta\rangle = \left(\begin{smallmatrix} z \\ t \end{smallmatrix}\right)$ *be two arbitrary single-qubit states. Compute explicitly the entries of vectors* $|\alpha\rangle \otimes |\beta\rangle$ *and* $|\beta\rangle \otimes |\alpha\rangle$*.*
3. *Show that it follows from the above definition of* SWAP *in eq. (1) that*

   $$\text{SWAP}\,|\alpha\rangle \otimes |\beta\rangle = |\beta\rangle \otimes |\alpha\rangle,$$

   for any single-qubit states $|\alpha\rangle, |\beta\rangle \in \mathbb{C}^2$*, not just the standard basis states.*

Exercise 2 (Controlled-NOT gate)**.** *The controlled-NOT gate, denoted* CNOT*, is a 2-qubit gate that flips the state of the second (target) qubit if the first (control) qubit is in the state* $|1\rangle$*. Its definition on the standard basis states and its graphical notation are as follows:*

$$\text{CNOT}\,|x,y\rangle = |x, y \oplus x\rangle, \qquad \begin{array}{ll} |x\rangle & |x\rangle \\ |y\rangle & |y \oplus x\rangle \end{array} \tag{2}$$



where $x, y \in \{0,1\}$*. The goal of this exercise is for you to understand how this gate works.*

1. *Write down* CNOT *as a* 4×4 *matrix.*
2. *Let* $X = \text{NOT} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ *be the Pauli X matrix that also implements the logical* NOT *operation. Verify that*

   $$\text{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X. \tag{3}$$

   What is the intuitive meaning of this identity?

3. *If we set $y = 0$ in eq. (2) then we get*

$$\mathrm{CNOT}\,|x,0\rangle = |x,x\rangle,$$

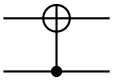
for any $x \in \{0,1\}$. Does this identity extend to arbitrary single-qubit quantum states (in a similar way to how eq. (1) generalized in case of the SWAP *gate)? That is, is it true that*

$$\mathrm{CNOT}\,|\psi\rangle \otimes |0\rangle = |\psi\rangle \otimes |\psi\rangle,$$

for any single-qubit state $|\psi\rangle \in \mathbb{C}^2$, not just the standard basis states? Prove this or find a counterexample.

Exercise 3 (Understanding the upside-down CNOT inside-out)**.** *The goal of this exercise is to derive the matrix representation for the upside-down* CNOT *gate*

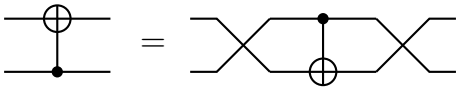
$$\text{CИOL} \quad = \quad$$



in three different ways.

1. *Write down how this gate acts on the four two-qubit computational basis states and derive its matrix representation from this.*

2. *Notice that*



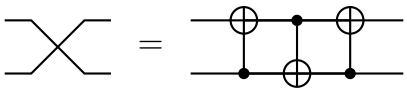
where crossed wires denote the SWAP *gate. In other words,*

$$\text{CИOL} = \mathrm{SWAP} \cdot \mathrm{CNOT} \cdot \mathrm{SWAP}\,.$$

Use the matrix representation of SWAP *from the previous exercise and the matrix representation of* CNOT *from the previous exercise to derive the matrix representation of* CИOL*.*

3. *What is the expression for* CИOL *that is analogous to eq. (3)? Compute the two terms in this expression and verify that they sum up to the correct matrix.*

Exercise 4. *Verify the following circuit identity:*



Also known as the XOR swap algorithm, it lets you exchange two variables without using additional memory:

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X := X XOR Y;
Y := X XOR Y;
X := X XOR Y;
```

Exercise 5 (Toffoli gate)**.** *The Toffoli gate is a three-qubit gate defined on the computational basis states as follows:*

$$|x_1, x_2, y\rangle \mapsto |x_1, x_2, y \oplus x_1 x_2\rangle,$$

for all $x_1, x_2, y \in \{0, 1\}$.

1. *Write down the Toffoli gate in matrix form.*
2. *Verify that the Toffoli gate is the same as the controlled-controlled-NOT gate, meaning that it flips the third (target) qubit if the first two (control) qubits are both in state* $|1\rangle$*. Write down the decomposition of the controlled-controlled-NOT gate analogous to eq. (3) and verify that you recover the Toffoli gate.*